

INTEGRATED MATH I

Chapter 1: Solving Linear Equations

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A. Warm-Up

1. $x + 2 = 7$

Hint: Subtract 2 from both sides.

Answer: 5

Solution:

$$x + 2 = 7 \rightarrow x = 5$$

2. $7x = 35$

Hint: Divide both sides by 7.

Answer: 5

Solution:

$$7x = 35 \rightarrow x = 5$$

3. $2x + 1 = 9$

Hint: Isolate the term with the variable, then divide.

Answer: 4

Solution:

$$2x + 1 = 9 \rightarrow 2x = 8 \rightarrow x = 4$$

B. Level Up

4. $3(x + 2) = 24$

Hint: Divide both sides by 3, then subtract 2.

Answer: 6

Solution:

$$3(x + 2) = 24 \rightarrow x + 2 = 8 \rightarrow x = 6$$

5. $3x + 3 = 5x + 0$

Hint: Collect x terms on one side, constants on the other, then divide.

Answer: 5

Solution:

$$3x + 3 = 5x + 0 \rightarrow (3-5)x = -3 \rightarrow x = 5$$

6. $(\frac{3}{3})x + 8 = 11$

Hint: Clear the fraction by multiplying both sides, then solve.

Answer: 3

Solution:

$$(3/3)x + 8 = 11 \rightarrow (3/3)x = 3 \rightarrow x = 3$$

C. Spot the Mistake

7. A student solved: $5x + 4 = 19$ $5x = 19$ $x = 19$ Where is the mistake?

Answer: 3

Solution:

Correct steps: $5x + 4 = 19 \rightarrow 5x = 15 \rightarrow x = 3$

D. Choose the Correct Move

(No items generated)

E. Match / Graph / Short Tasks

(No items generated)

F. Challenge

8. Solve: $0.6x + 0 = 6.0$

Hint: Isolate and divide by the coefficient (consider decimals).

Answer: 10

Solution:

Solve: $0.6x + 0 = 6.0 \rightarrow 0.6x = 6.0 \rightarrow x = 10$